Reliable Machine Learning with Unreliable Black Boxes **Spring 2026**

Lecture 17, part 2: Other Topics

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This lecture surveys several frameworks for learning that remain reliable when the test distribution may differ from the training distribution. We begin with *testable learning under distribution shift*, contrast it with *PQ learning*, and then turn to *distributionally robust optimization* and its specialization to *adversarial and certified robustness*.

1 Testable Learning with Distribution Shift

Consider learning under *covariate shift*. We have a training distribution P and a test distribution Q; the learner sees labeled samples from P together with *unlabeled* samples from Q,

$$S_P = \{(x_i, y_i)\}_{i=1}^{n} \sim P, \qquad T_Q = \{x'_j\}_{j=1}^{m} \sim Q_X,$$

where Q_X is the marginal of Q on the instance space. The covariates may shift between training and test time, but test labels are never observed.

The difficulty is plain: without labels from Q, there is no direct way to measure test error. *Testable learning with distribution shift* (TDS learning) [Klivans, Stavropoulos, and Vasilyan, 2024] sidesteps this by letting the algorithm decline. Rather than always returning a classifier, the algorithm $A(S_P, T_Q)$ may instead report that the test data look too different from the training data to be trusted:

$$A(S_P, T_Q) \in \{(, h) : h \in \mathcal{H}\} \cup \{\}.$$

It is required to satisfy two properties.

Soundness. Whenever A accepts and outputs h, that h has small error on the test distribution. In the realizable setting this means ${}_Q(h) \leq \varepsilon$; in the agnostic setting it means competitiveness with the best classifier on Q,

$${}_Q(h) \leq_Q (\mathcal{H}) + \varepsilon,$$

possibly with a benchmark depending on both P and Q. Acceptance thus serves as a *certificate* of good test performance.

Completeness. When there is no shift, A accepts with high probability. At the level of instance marginals,

$$Q_X = P_X \quad \Longrightarrow \quad \Pr[A(S_P, T_Q) =] \geq 1 - \delta.$$

These two requirements give rejection a clean reading. By completeness, $Q = P$ forces the rejection probability to be at most δ, so observing a rejection is—except with probability δ—statistical evidence that $Q \neq P$. Rejection is therefore not a deterministic proof of shift, but a calibrated warning

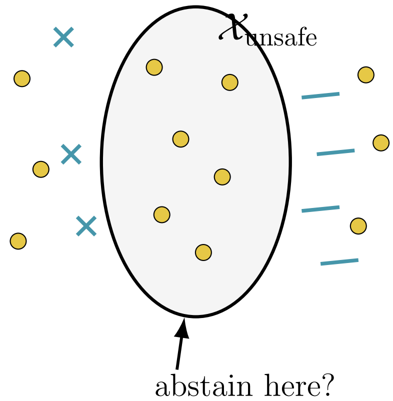

Figure 1: A schematic comparison with PQ learning. A PQ learner may abstain on a region of the instance space, whereas a TDS learner accepts or rejects the entire test distribution.

that the test distribution cannot safely be treated as identical to the training one. Symmetrically, acceptance certifies low test error, while rejection simply withholds that certificate.

2 TDS Learning versus PQ Learning

TDS learning is close in spirit to *PQ learning*: both refuse to predict blindly everywhere. The frameworks differ in *where* that refusal is made.

Global versus local rejection. TDS learning makes a single decision about the whole test distribution. If it accepts, the returned h is guaranteed on all of Q; if it rejects, it discards Q entirely. The certificate is *distribution-level*, and the algorithm never decides point by point.

PQ learning refuses more locally. It outputs a classifier together with a region $\mathcal{X}_{\text{safe}} \subseteq \mathcal{X}$ on which it is willing to commit: it predicts $h(x)$ for $x \in \mathcal{X}_{\text{safe}}$ and abstains otherwise (Figure 1). The certificate is *pointwise*—valid only on the region the learner chooses to predict on.

Assumptions on P_X. The two frameworks also assume different structure on the training marginal. TDS results typically take P_X to be known or to lie in a nice family—for instance $P_X = \mathcal{N}(0, I)$, or more generally a Gaussian or log-concave distribution. This structure is what buys efficiency: when P_X is Gaussian, the geometry of linear classifiers $h_w(x) = \text{sign}(\langle w, x \rangle)$ is tractable, the learner can test whether the unlabeled Q_X samples are compatible with the classifier it has learned, and one obtains polynomial-time guarantees—even agnostically, with the goal $_Q(h) \leq_Q (\mathcal{H}) + \varepsilon$.

PQ learning, by contrast, is distribution-free in P_X: it targets arbitrary marginals. The price of that generality is computational. Lacking geometric structure to exploit, PQ learning usually assumes stronger algorithmic primitives, such as an ERM oracle or a reliable learning oracle. The contrast is summarized below.

	TDS learning	**PQ learning**
Reject option	rejects the whole test distribution	abstains on points or regions
Certificate	distribution-level	local / pointwise
Assumption on P_X	known or nice family (e.g. Gaussian)	arbitrary P_X
Computation	poly-time in structured settings	ERM / reliable-oracle assumptions

3 Distributionally Robust Optimization

Standard supervised learning minimizes the population risk under P,

$$\min_{h\in\mathcal{H}} \mathbb{E}_{(x,y)\sim P}\big[\ell(h(x),y)\big],$$

which is a good proxy for test performance only when the test distribution resembles P. Under shift this can fail badly: a classifier with small loss on P may do poorly on a shifted Q.

Distributionally robust optimization (DRO) [Duchi and Namkoong, 2021] hedges against this by minimizing the worst-case risk over a neighborhood of P,

$$\min_{h\in\mathcal{H}} \sup_{\widehat{P}:\, d(\widehat{P},P)\le\varepsilon} \mathbb{E}_{(x,y)\sim\widehat{P}}\big[\ell(h(x),y)\big],$$

a minimax game in which the learner chooses h and an adversary chooses the worst admissible $\widehat{P}$. Here $d(\widehat{P},P)$ is a divergence between distributions and ε sets the radius of the uncertainty set. Where ERM asks which classifier is best on P, DRO asks which is best against the worst distribution *near* P—a more conservative criterion that penalizes classifiers whose low average loss conceals fragility under reweighting.

Choice of divergence. The neighborhood is determined by $d(\widehat{P},P)$. Common choices—f-divergences (including the χ^2- and KL-divergences) and the Wasserstein distance—encode different notions of robustness. An f-divergence limits how much $\widehat{P}$ may reweight the mass of P, whereas the Wasserstein distance limits how far probability mass must be transported in the instance space.

f-divergence sets and Fenchel duality. For an f-divergence the uncertainty set is

$$\mathcal{U}_\varepsilon(P) = \big\{\widehat{P}:\ D_f(\widehat{P}\,\|\,P)\le\varepsilon\big\},$$

and the inner supremum lets the adversary upweight high-loss examples subject to the divergence budget. Crucially, Fenchel duality rewrites this infinite-dimensional maximization over distributions as an optimization over a few scalar dual variables, turning the robust risk into a computable, regularized loss. DRO is therefore not merely conceptual: for suitable divergences it reduces to a tractable objective.

Example: χ^2-DRO and variance regularization. For the χ^2 uncertainty set $D_{\chi^2}(\widehat{P} \,\|\, P) \le \varepsilon$, the robust risk becomes (informally) a mean–variance objective,

$$\mathbb{E}_P[\ell(h)] + \lambda\sqrt{{}_P(\ell(h))}, \qquad \ell(h) = \ell(h(x), y).$$

Variance regularization is exactly the right penalty here: a classifier with low average but highly variable loss is precisely the one an adversary can exploit by reweighting toward its high-loss region.

Relation to TDS learning. Both DRO and TDS learning address shift, but they commit differently. DRO fixes an uncertainty set in advance and always returns a classifier robust to every $\widehat{P} \in \mathcal{U}_\varepsilon(P)$, without ever seeing the actual test data. TDS learning instead inspects unlabeled samples from the real Q and then either certifies a classifier on Q or rejects. In short, DRO optimizes for a worst case it postulates, while TDS learning tests the case it is actually given.

4 Adversarial and Certified Robustness

Adversarial robustness is the special case of DRO in which the uncertainty arises from local, per-example perturbations. Instead of asking only for accuracy at x, we ask for accuracy throughout a small ball $B(x, \varepsilon)$ (typically under ℓ_2 or ℓ_∞):

$$\min_{h \in \mathcal{H}} \mathbb{E}_{(x,y)\sim P}\Big[\sup_{\widetilde{x} \in B(x,\varepsilon)} \ell(h(\widetilde{x}), y)\Big].$$

The inner supremum is the adversary, perturbing each example to maximize loss; the outer minimization seeks a classifier robust to those perturbations. Because the adversary moves mass only locally rather than reweighting globally, this is a Wasserstein-type DRO—specifically a W_∞ constraint $W_\infty(\widehat{P}, P) \le \varepsilon$ when every point may move by at most ε.

Certified robustness. Adversarial training makes a classifier empirically robust but does not prove that *every* perturbation in the ball is safe. *Certified* robustness supplies that proof: alongside the label $h(x)$ it returns a radius $r(x)$ with the guarantee

$$\|\widetilde{x} - x\| \le r(x) \quad \Longrightarrow \quad h(\widetilde{x}) = h(x).$$

No perturbation inside $B(x, r(x))$ can change the prediction—an explicit mathematical certificate rather than an empirical one.

Randomized smoothing. A standard route to such certificates is *randomized smoothing* [Cohen, Rosenfeld, and Kolter, 2019]. From any base classifier h, define the smoothed classifier that returns the most likely label under Gaussian input noise $\eta \sim \mathcal{N}(0, \sigma^2 I)$:

$$g(x) = \arg\max_y \Pr_{\eta\sim\mathcal{N}(0,\sigma^2 I)}\big[h(x+\eta) = y\big].$$

Equivalently, classify many noisy copies of x and take the majority vote. When one label is sufficiently more likely than the others under the noise, g provably does not change on a small ℓ_2-ball—converting an arbitrary base classifier into one with certified ℓ_2 robustness.

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